WEEK 7

1) A Distributed Hash Table (DHT) supports operations like insert/lookup/delete in a:
A. Single-machine setting only
B. Distributed setting using keys
C. Only SQL setting
D. Only centralized setting

2) In Chord, a key K is stored at:
A. The peer with the smallest ID
B. The first peer with ID ≥ K (mod 2^m)
C. The peer whose ID equals K always
D. Any random peer for load balancing

3) Virtual nodes are used primarily to:
A. Reduce disk latency
B. Prevent node joins
C. Enforce strong consistency
D. Improve load balancing by reducing variance

4) The map() function outputs:
A. Intermediate key/value pairs
B. Final sorted output files
C. Only a single global value
D. Only metadata pointers

5) The reduce() function primarily:
A. Splits files into chunks
B. Merges all intermediate values associated with the same key
C. Performs DNS lookup
D. Encrypts intermediate data

6) In MapReduce, shuffle refers to:
A. Transfer of intermediate map outputs to reducers based on key partitioning
B. CPU scheduling inside a node
C. Encrypting data at rest
D. Replicating HDFS blocks

7) In YARN, the component responsible for cluster-wide scheduling is the:
A. Node Manager (NM)
B. Tracker
C. Application Master (AM)
D. Resource Manager (RM)

8) The Application Master (AM) mainly handles:
A. Only disk replication
B. Container negotiation and task failure handling for its job
C. DNS resolution
D. Maintaining the global file system namespace

9) “Stragglers” in MapReduce are:
A. Tasks that finish first
B. Slow tasks that delay the job completion
C. Failed tasks only
D. Reducers that never shuffle

10) A combiner is beneficial when the reduce operation is:
A. Non-deterministic
B. Dependent on global ordering only
C. Commutative and associative, enabling local partial aggregation
D. Only for encryption
Answer: C